

Qimin Yan

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Professional Experience

2022- present Associate Professor,
Department of Physics, Northeastern University
2016- 2022 Assistant Professor,
Department of Physics, Temple University
2013-2016 Post-doctoral researcher,
Molecular Foundry, Lawrence Berkeley National Laboratory
Department of Physics, University of California, Berkeley
2012 Post-doctoral researcher,
Materials Department, University of California, Santa Barbara

Leadership and Professional Service

- Editorial Board Member, Journal of AI-Driven Materials Research, 2026–present
- Organizer, Principles and Applications of Symmetry in Magnetism (PASM) Summer School, (July 2026)
- Guest Editor, “Roadmap: 2D Materials for Quantum Technologies”, 2D Materials (2025)
- Session Chair, “Integrating Machine Learning and Simulations for Materials Modeling”, MRS Spring Meeting (2023)
- Co-organizer, APS March Meeting Focus Topic, “Computational Discovery and Design of Novel Materials” (2017)
- Symposium Organizer, MRS Spring Meeting, “2D Materials Theory, Discovery and Design” (2019)
- Proposal reviewer for NSF (AI Institute, CAREER, CDS&E, OAC SSE, DMR CMP) and DOE (BES Early Career, CMS, QIS, EPSCoR, Data Science Program).
- Proposal Review Board Member for the Center for Nanoscale Materials at Argonne National Laboratory.
- Referee of Nature, Nature Chemistry, Physical Review Letters, Nature Communications, ACS Nano, Materials Today, Nano Letters, npj Computational Materials, Chemical Communications, Chemistry of Materials, Physical Chemistry Chemical Physics, Applied Physics Letters, Journal of Applied Physics, Journal of Physics: Condensed Matter, Journal of Materials Chemistry A, Computational Materials Science, Journal of Nanomaterials, etc.

Awards and Fellowships

2022 NSF CAREER Award, National Science Foundation
2021 Selected as “Emerging Leaders 2021” by Journal of Physics: Condensed Matter.
2020 Finalist for “2020 Rising Stars in Computational Materials Science Prize”
2019 DOE Early Career Award, U.S. Department of Energy

- 2017 NERSC Award for High Impact Scientific Achievement
- 2011 Outstanding Graduate Student Research Achievement Award,
Solid-State Lighting and Energy Center, University of California, Santa Barbara.

Research Funding

2023–2027 National Science Foundation (NSF), Designing Materials to Revolutionize and Engineer our Future (DMREF) Program, \$440,000, “DMREF: Deep learning guided twistronics for self-assembled quantum optoelectronics”. (PI at Northeastern)

2023-2028 National Science Foundation (NSF), ExpandQISE Program, \$1,350,000, “A Quantum Science Education and Research Program for HBCUs: Exotic Physics and Applications of Solid-State Qubits”. (senior investigator at Northeastern)

2022 – 2027 National Science Foundation (NSF), Faculty Early Career Development Program (CAREER) Program, \$507,958, “CAREER: Quantum defects in two-dimensional materials by local-symmetry-guided data-driven design”. (PI)

2023 – 2026 Department of Energy (DOE), Early Career Research Program, \$750,000, “Synthesis of motif and symmetry for accelerated learning, discovery, and design of electronic structures for energy conversion applications”. (PI)

2019 – 2022 Department of Energy (DOE), Early Career Research Program, \$750,000, “Synthesis of motif and symmetry for accelerated learning, discovery, and design of electronic structures for energy conversion applications”. (PI)

2018 – 2022 Department of Energy (DOE), Quantum Information Science Program, \$405,226, “Design, Control and Application of Next-Generation Qubits”. (PI at Temple)

2018 – 2020 Department of Energy (DOE), EFRC Center, \$110,582, “Center for Complex Materials from First Principles (CCM)”. (co-PI)

2016 – 2018 Department of Energy (DOE), EFRC Center, \$200,000, “Center for the Computational Design of Functional Layered Materials (CCDM)”. (co-PI)

Invited Presentations

Extensive invited presentations at international conferences (APS March Meeting, MRS Spring and Fall Meeting, ICDS Workshop, Gordon Research Conference), leading universities and national labs.

1. Colloquium, Department of Physics and Astronomy, University of Kansas, Lawrence, KS (October 2025)

From simple to complex: learning pathways for solid-state quantum systems

2. Northeast Quantum Forum 2025, Durham, NH (October 2025)

Learning pathways for solid-state quantum systems: electronic structure and configurational properties

3. MRS Spring Meeting, Seattle, WA (April 2025)

Symmetry-guided and data-driven discovery of quantum defects in two-dimensional materials

4. APS Global Physics Summit, Anaheim, CA (April 2025)

Symmetry-guided and AI-assisted design of quantum defects in two-dimensional materials

5. MRS Fall Meeting, Boston, MA (December 2024)

- Data-driven and machine learning assisted design of functional defects in two-dimensional materials*
6. Energy Science Institute Seminar, Yale University, New Haven, CT (October 2024)
Physical principle informed machine learning for complex materials in quantum regime
 7. GRC Conference for Defects in Semiconductors, Newry, MN (August 2024)
Symmetry-guided and machine learning assisted design of quantum defects in 2D materials
 8. AI Retreat, University of Alabama, Birmingham, AL (January 2024)
Physical principle informed AI and machine learning for science and engineering problems
 9. Condensed Matter Seminar, Boston University, Boston, MA (November 2023)
Data-driven materials design in the quantum regime
 10. NST Colloquium, Argonne National Laboratory, Chicago, IL (October 2023)
Machine learning in the quantum regime through physical-principle-informed representations
 11. International Conference on Defects in Semiconductors, Rehoboth Beach, DE (Sept 2023)
Data-driven discovery and design of quantum defects in 2D materials
 12. 50 years of DFT, Celebration of John Perdew's 80th birthday, New Orleans, LA (June 2023)
Incorporation of density scaling constraint in density functional design via contrastive learning
 13. Materials-2023 workshop, Houston, Texas (April 2023)
Symmetry-enabled data-driven design of quantum defects in 2D materials
 14. MRS Spring Meeting, San Francisco, CA (April 2023)
Machine learning in the quantum regime through physical-principle-informed representations
 15. PCSI-48 workshop, Redondo Beach, CA (January 2023)
Machine learning in the quantum regime through physical-principle-informed representations
 16. Condensed Matter Seminar, University of Texas, Austin, TX (November 2022)
Machine learning in the quantum regime through physical-principle-enabled representation learning
 17. Joint Colloquium, Colleges of Science and Engineering, Northeastern University, MA (April 2022)
Data-driven materials design in the quantum regime
 18. Materials-2022 workshop, Newton, MA (April 2022)
Quantum defects in two dimensional materials: local-symmetry-guided discovery and design
 19. Department Colloquium, Department of Physics, University of Florida, FL (Jan. 2022)
Data-driven materials design in the quantum regime: motif centric learning and symmetry-guided materials discovery
 20. Condensed Matter Seminar, Stony Brook University, NY (2021)
Data-driven materials design in the quantum regime
 21. Department of Materials Science and Engineering, University of Pennsylvania, PA (2021)

Data-driven materials design in the quantum regime

22. Physics Colloquium, Temple University, Philadelphia, PA (2020)

Symmetry enabled effective learning and accelerated discovery of quantum materials

23. SCAN Workshop, Temple University, Philadelphia, PA (2019)

Data-driven discovery of functional 2D materials using a SCAN-enabled electronic structure database

24. Department of Physics, Penn State University, University Park, PA (2018)

Data-driven discovery of functional 2D materials utilizing a 2D electronic structure database

25. Lecture Series on Materials Theory and Computation, Xi'an, China (2018)

Data-driven discovery of functional 2D materials

26. Department of Applied Physics & Materials, California Institute of Technology, CA (2018)

Data-driven discovery of functional 2D materials utilizing a 2D electronic structure database

27. International Conference on Low-dimensional Quantum Materials, Snowbird, UT (2018)

Data-driven discovery of functional 2D materials utilizing a 2D electronic structure database

28. MSE Colloquium, Boston University, MA (2017)

Discovery of functional energy and topological materials with a combination of high-throughput theory and experiment

29. College of Materials Science & Engineering, Jilin University, Changchun, China (2017)

Data-driven discovery of solar fuels photoanode materials

30. Institute of Physics, Chinese Academy of Science, China (2017)

Discovery of solar fuels photoanode materials with a combination of high-throughput theory and experiment

31. Department of Physics, Xi'an Jiaotong University, Xi'an, China (2017)

Discovery of solar fuels photoanode materials with a combination of high-throughput theory and experiment

32. Hefei National Laboratory for Physical Sciences at the Microscale, USTC, Hefei, China (2017)

Discovery of solar fuels photoanode materials with a combination of high-throughput theory and experiment

33. Beijing Computational Science Research Center, Beijing, China (2017)

Materials Genome Initiative and data-driven discovery of solar fuels photoanode materials

34. Electronic Materials and Applications 2017, Orlando, FL (2017)

Discovery of solar fuels photoanode materials by integrating high-throughput theory and experiment

35. Department of Materials Science and Engineering, University of Michigan, Ann Arbor, MI (2016)

First-principles data-driven discovery of transition metal oxides for artificial photosynthesis

36. Department of Materials Engineering, Purdue University, West Lafayette, IN (2016)

Predictive design of functional materials and devices: from high-throughput computation to multi-scale modeling

37. Department of Materials Science and Engineering, University of California, Riverside, CA (2016)
Discovery and design of functional materials and devices: from high-throughput calculations to multi-scale modeling
38. American Physical Society March Meeting 2016, Baltimore, MD (2016)
First-principles data-driven discovery of transition metal oxides for artificial photosynthesis
39. Department of Physics & Astronomy, Rutgers University, Piscataway, NJ (2016)
Discovery and design of complex materials with ab initio high-throughput approaches
40. Department of Materials Science and Engineering, University of Delaware, DE (2015)
Computational design of new materials for energy applications: from high-throughput calculations to multi-scale modeling
41. Department of Physics, Xi'an Jiaotong University, China (2015)
Materials genome initiative and data-driven discovery of novel energy and topological materials
42. Department of Materials Science and Engineering, Shanghai Jiaotong University, China (2015)
Materials genome initiative and data-driven discovery of novel energy and topological materials
43. Materials Genome Institute, Shanghai University, China (2015)
Materials genome initiative and data-driven discovery of novel energy and topological materials
44. School of Physics, Nankai University, China (2015)
Materials genome initiative and data-driven discovery of novel energy and topological materials
45. Institute of Physics, Chinese Academy of Science, China (2015)
Materials genome initiative and data-driven discovery of novel energy and topological materials
46. Division of Energy & Environment, Shenzhen Graduate School of Tsinghua University, China (2015)
Materials genome initiative and data-driven discovery of novel energy and topological materials
47. Department of Materials Science and Engineering, Tsinghua University, China (2015)
Materials genome initiative and data-driven discovery of novel energy and topological materials
48. Materials Department, University of California, Santa Barbara, CA (2014)
Ab Initio high-throughput approach for discovery of stable transition metal oxides for solar energy capture and conversion
49. Molecular Foundry, Lawrence Berkeley National Laboratory, CA (2012)
Theoretical Study of Material and Device Properties of Group-III Nitrides
50. Department of Physics, Tsinghua University, China (2011)
The role of nitrogen vacancies and related complexes in the luminescence and p-type doping compensation of GaN

Education

- 2007-2012 **Ph. D in Materials**, Materials Department, University of California, Santa Barbara
Supervisor: Prof. Chris G. Van de Walle and Prof. Matthias Scheffler
Thesis: Theoretical Study of Material and Device Properties of Group-III Nitrides

2003-2006 **M. S. in Physics**, Department of Physics, Tsinghua University, China
 1999-2003 **B. S. in Applied Physics**, Dept. of Applied Physics, Xi'an Jiaotong University, China

Teaching Experience

Northeastern University

Spring 2026 Physics 4211 – Artificial Intelligence in Physical Science (New course developed)
 Spring 2025 Physics 2303 – Modern Physics
 Spring 2024 Physics 2303 – Modern Physics
 Spring 2023 Physics 2303 – Modern Physics

Temple University

Spring 2022 Physics 4511 - Scientific Computing III
 Fall 2021 Physics 1062 - Elementary Classical Physics II
 Spring 2021 Physics 2511 - Scientific Computing I
 Physics 4511 - Scientific Computing III
 Fall 2020 Physics 1062 - Elementary Classical Physics II
 Spring 2020 Physics 4511 - Scientific Computing III
 Fall 2019 Physics 8702 - Solid State Physics
 Spring 2019 Physics 8702 - Solid State Physics
 Fall 2018 Physics 2021 - General Physics
 Spring 2018 Physics 8702 - Solid State Physics
 Fall 2017 Physics 1061 - Elementary Classical Physics I
 Fall 2016 Physics 1061 - Elementary Classical Physics I

Mentoring Experience

- Associate Research Scientist at Northeastern: Yubo Qi (2022-2023)
- Postdoc Researcher at Northeastern: Zhenyao Fang (2023–present); Jeng-Yuan Tsai (2022-2023)
- Graduate Students at Northeastern: Weiyi Gong (PhD in 2025; now Postdoctoral Fellow at Argonne National Laboratory), Ongoing: Ting-Wei Hsu, Anoj Aryal, Alex Heilman, Yu Ruan
- Trainees placed in faculty positions: Yubo Qi at University of Alabama, Birmingham; Liping Yu at University of Central Florida; Jinbo Pan at Chinese Academy of Sciences.

Northeastern University

2025-present Yu Ruan, Graduate Student
 2024-present Ting-Wei Hsu, Graduate Student
 2023-present Zhenyao Fang, Postdoc Researcher
 2022-2026 Anoj Aryal, Graduate Student
 Alex Heilman, Graduate Student
 2022-2025 Weiyi Gong, Graduate Student
 (Now as a Postdoctoral Fellow at Argonne National Laboratory)
 2023-2024 Claire Schlesinger, Undergraduate Co-Op student

2022-2023 Jeng-Yuan Tsai, Postdoc Researcher
(Now as a Data Processing Scientist at Jefferson Lab)
Yubo Qi, Associate Research Scientist
(Now as an Assistant Professor at University of Alabama, Birmingham)

Temple University

2021-2022 Anoj Aryal, Graduate Student
Alex Heilman, Graduate Student
2021-2022 Debajit Chakraborty, Research Assistant Professor
2020-2022 Andy Philips, Undergraduate Student
2020-2021 Brendan Magdamo, Undergraduate Student
2019-2022 Weiyi Gong, Graduate Student
2019-2020 Lifu Zhang, Visiting Student
2019 summer Francesco Ricci, Visiting Scholar
2018-2022 Jeng-Yuan Tsai, Graduate Student
2018-2020 Yijun Tong, Visiting Master Student
(Now as a PhD student at University of Washington)
2017 Linh Nguyen, Undergraduate Student
2017 Dylan Harrison, Undergraduate Student
2016-2020 Huta R Banjade, Graduate Student
(Now as a Senior Research Scientist at Badger Meter)
Yu Wang, Undergraduate Student
(Now as a PhD student at Princeton University)
2016-2019 Jinbo Pan, Postdoc Fellow
(Now as an Associate Professor at Chinese Academy of Sciences)
Yanfang Zhang, Visiting Scholar
(Now as a Postdoc Fellow at Chinese Academy of Sciences)
2016-2018 Liping Yu, Research Assistant Professor
(Now as an Associate Professor at University of Central Florida)

Publications

108 scientific publications; **2** U.S. patents; **1** book chapter.

Google Scholar link: <https://scholar.google.com/citations?user=ysfTfdkAAAAJ&hl=en>

Total citations: **8,526**; h-index: **45**; **22** publications with over 100 citations.

- 108 T.-W. Hsu, A. Bansil, **Q. Yan**, “Symmetry- and Property-Aware Crystal Generation with Reinforcement Learning for Inverse Materials Design”, arXiv:2609.13468 (2026), DOI: 10.48550/arXiv.2609.13468
- 107 A. J. Heilman, **Q. Yan**, “Point Group Equivariant Graph Neural Networks for Materials”, arXiv:2607.16871 (2026), DOI: 10.48550/arXiv.2607.16871
- 106 Z. Fang, **Q. Yan**, “A Multi-Scale Machine Learning Framework for Coupled Chemical, Spin, and Structural Disorder in Alloys”, arXiv:2607.07456 (2026), DOI: 10.48550/arXiv.2607.07456
- 105 Z. Wang, Y. Hu, Z. Fang, Y. Wang, P. Yan, J. Sun, M. Lopez, R. Stadel, Y. Niu, J. M. Little, **Q. Yan**, P.-Y. Chen, C. Gong, J. W. Bennett, G. Gu, Q. Li, Z. Chen, T. J. Woehl, E. E. Rodriguez, K. Liu, S. Ren, “Chiral Molecular Intercalation Enables Light-Controlled 2D Multiferroic Heterostructures”, *Nano Letters* 26, 6883–6890 (2026), DOI: 10.1021/acs.nanolett.6c00485
- 104 X. Zhang, M. E. Turiansky, L. Razinkovas, M. Maciaszek, P. Broqvist, **Q. Yan**, J. L. Lyons, C. E. Dreyer, D. Wickramaratne, Á. Gali, A. Pasquarello, C. G. Van de Walle, “First-principles calculations of defects and electron–phonon interactions: Seminal contributions of Audrius Alkauskas to the understanding of recombination processes”, *Journal of Applied Physics* 135, 150901 (2024), DOI: 10.1063/5.0205525
- 103 W. Gong, T. Sun, H. Bai, J.-Y. Tsai, H. Ling, **Q. Yan**, “Graph Transformer Networks for Accurate Band Structure Prediction: An End-to-End Approach”, *submitted* (2026) DOI: 10.48550/arXiv.2411.16483
- 102 A. Aryal, W. Gong, H. Banjade, **Q. Yan**, “Materials Representation Learning Based on a Material–Motif Network and Heterogeneous Graphs”, *Advanced Intelligent Discovery*, e70142 (2026), DOI: 10.1002/aidi.70142
- 101 A. Heilman, C. Schlesinger, **Q. Yan**, “Equivariant Graph Neural Networks for Prediction of Tensor Material Properties of Crystals”, *submitted* (2026), DOI: 10.48550/arXiv.2406.03563
- 100 J. Y. Tsai, W. Gong, **Q. Yan**, “Symmetry-guided data-driven discovery of native quantum defects in two-dimensional materials”, *submitted* (2026), DOI: 10.48550/arXiv.2405.11379
- 99 Y. Liu, A. Aryal, K. Yang, D. Calugaru, Z. Fang, H. Hu, **Q. Yan**, B. A. Bernevig, C. Liu, “Ideal Topological Flat Bands in Two-dimensional Moiré Heterostructures with Type-II Band Alignment”, *submitted* (2026), DOI: 10.48550/arXiv.2507.06168
- 98 Z. Fang, **Q. Yan**, “Towards the Predictive Design of Quantum Defects for Next-Generation Quantum Technologies”, *Communications Materials* 7, 155 (2026), DOI: 10.1038/s43246-026-01225-7

- 97 **Q. Yan**, ... “Roadmap: 2D Materials for Quantum Technologies”, *2D Materials* *accepted* (2026) DOI: 10.48550/arXiv.2512.14973
- 96 T.-W. Hsu, Z. Fang, A. Bansil, **Q. Yan**, “Accurate Prediction of Tensorial Spectra Using Equivariant Graph Neural Network”, *Nature Communications* *17*, 3330 (2026), DOI: 10.1038/s41467-026-69159-9
- 95 Z. Fang, T.-W. Hsu, **Q. Yan**, “Learning Atomic Representations for Data-Driven Materials Design”, *AI for Science* *2*, 013001 (2026), DOI: 10.1088/3050-287x/ae45bf
- 94 A. Heilman, W. Gong, **Q. Yan**, “Crystal Hypergraph Convolutional Networks”, *npj Computational Materials* *11*, 336 (2025), DOI: 10.1038/s41524-025-01826-9
- 93 Z. Fang, T.-W. Hsu, **Q. Yan**, “A Machine Learning Framework for Modeling Ensemble Properties of Atomically Disordered Materials”, *ACS Nano* *19*, 37353 (2025), DOI: 10.1021/acsnano.5c13080
- 92 S. Vaidya, X. Gao, S. Dikshit, Z. Fang, A. E. Llacsahuanga Allica, Y. P. Chen, **Q. Yan**, T. Li, “Coherent Spins in van der Waals Semiconductor GeS₂ at Ambient Conditions”, *Nano Letters* *25*, 14356 (2025), DOI: 10.1021/acs.nanolett.5c03567
- 91 L. Frammolino, M. Holbrook, C. Lei, J.-Y. Tsai, Y. Wan, L.Y. Huang, L.-J. Li, **Q. Yan**, A. MacDonald, C.-K. Shih, “Microscopic Investigations of Point Defect Interactions in WS₂ Monolayers”, *ACS Nano* *19*, 33059-33069 (2025), DOI: 10.1021/acsnano.4c09204
- 90 Z. Fang, T.-W. Hsu, **Q. Yan**, “Dataset of tensorial optical and transport properties of materials from the Wannier function method”, *Scientific Data* *12*, 1092 (2025), DOI: 10.1038/s41597-025-05396-9
- 89 Y. Qi, W. Gong, **Q. Yan**, “Bridging deep learning force fields and electronic structures with a physics-informed approach”, *npj Computational Materials* *11*, 177 (2025), DOI: 10.1038/s41524-025-01668-5
- 88 **Q. Yan**, J. L. Lyons, L. Gordon, A. Janotti, C. G. Van de Walle, “Oxygen impurities in AlN and their impact on optical absorption” *Applied Physics Letters* *126*, 062106 (2025), DOI: 10.1063/5.0234655
- 87 Z. Fang, **Q. Yan**, “Leveraging Persistent Homology Features for Accurate Defect Formation Energy Predictions via Graph Neural Networks”, *Chemistry of Materials* *37*, 1531–1540 (2025), DOI: 10.1021/acs.chemmater.4c03028
- 86 Z. Wang, T. Xie, Z. Fang, J. Zhang, C. Gong, **Q. Yan**, S. Ren, “Chiral molecular magnet superstructures with light control”, *Nano Letters* *25*, 2502-2508 (2025), DOI: 10.1021/acs.nanolett.4c06107
- 85 Z. Fang, **Q. Yan**, “Towards accurate prediction of configurational disorder properties in materials using graph neural networks”, *npj Computational Materials* *10*, 91 (2024), DOI: 10.1038/s41524-024-01283-w
- 84 Z. Wang, Y. Huang, W. Gong, **Q. Yan**, S. Ren, “Lithiation bridged molecular conducting magnets”, *Applied Materials Today* *38*, 102188 (2024), DOI: 10.1016/j.apmt.2024.102188

- 83 Z. Wang, Q. Wang, W. Gong, A. Chen, A. Islam, L. Quan, T. J. Woehl, Q. Yan, S. Ren, "Magnet-in-ferroelectric crystals exhibiting photomultiferroicity", *PNAS* 121, e2322361121 (2024), DOI: 10.1073/pnas.2322361121
- 82 G. Kim, B. Huet, C. E. Stevens, K. Jo, J.-Y. Tsai, S. Bachu, M. Leger, S. Song, M. Rahaman, K. Y. Ma, N. R. Glavin, H. S. Shin, N. Alem, **Q. Yan**, J. R. Hendrickson, J. M. Redwing, D. Jariwala, "Confinement of excited states in two-dimensional, in-plane, quantum heterostructures", *Nature Communications* 15, 6361 (2024), DOI: 10.1038/s41467-024-50653-x
- 81 **Q. Yan**, S. Chowdhury, A. Bansil, S. Kar, "The Case for a Defect Genome Initiative", *Adv. Mater.* 36, 2303098 (2024). DOI: 10.1002/adma.202303098
- 80 Q. Gao, **Q. Yan**, Z. Hu, L. Chen "Bilayer Kagome Borophene with Multiple van Hove Singularities", *Adv. Sci.* 11, 2305059 (2024), DOI: 10.1002/advs.202305059
- 79 W. Gong, T. Sun, H. Bai, S. T. u. R. Chowdhury, P. Chu, A. Aryal, J. Yu, H. Ling, J. P. Perdew, **Q. Yan**, "Incorporation of density scaling constraint in density functional design via contrastive representation learning", *Digital Discovery* 2, 1404 (2023). DOI: 10.1039/D3DD00114H
- 78 Y. Huang, A. Pathak, C. Rumsey, N. Kramer, Y. Hu, M. Trebbin, J.-Y. Tsai, M. Ivill, **Q. Yan**, S. Ren, "Pressure-Controlled Magnetism in 2D Molecular Layers", *Nature Commun.* 14, 3186 (2023). DOI: 10.1038/s41467-023-38991-8
- 77 S. Singh, W. Gong, C. Stevens, J. Hou, A. Singh, H. Zhang, S. B. Anantharaman, A. Mohite, J. Hendrickson, **Q. Yan**, D. Jariwala, "Valley Polarized Interlayer Excitons in 2D Chalcogenide-Halide Perovskite-Van der Waals Heterostructures", *ACS Nano* 17, 7487 (2023), DOI: 10.1021/acsnano.2c12546
- 76 C. Becher, W. Gao, S. Kar, C. Marciniak, T. Monz, J. G. Bartholomew, P. Goldner, H. Loh, E. Marcellina, K. E. J. Goh, T. S. Koh, B. Weber, Z. Mu, J.-Y. Tsai, **Q. Yan**, T. Huber-Loyola, S. Höfling, S. Gyger, S. Steinhauer, V. Zwiller, "2023 Roadmap for Materials for Quantum Technologies", *Mater. Quantum. Technol.* 3, 012501 (2023), DOI: 10.1088/2633-4356/aca3f2
- 75 H. Tang, S. Neupane, **Q. Yan**, A. Ruzsinszky, "Density Functional Theory Study of Controllable Optical Absorptions and Magneto-Optical Properties of Magnetic CrI₃ Nanoribbons: Implications for Compact 2D Magnetic Devices", *ACS Appl. Nano Mater.* 5, 14388 (2022), DOI: 10.1021/acsanm.2c02722
- 74 Y. Huang, W. Gong, G. Zhang, Z. Li, H. Lin, **Q. Yan**, S. Ren, "Dimensional Transformation of Molecular Magnetic Materials" *ACS Nano* 16, 13232 (2022), DOI: 10.1021/acsnano.2c06912
- 73 J.-Y. Tsai, J. Pan, H. Lin, A. Bansil, **Q. Yan**, "Antisite defect qubits in monolayer transition metal dichalcogenides", *Nat. Commun.* 13, 492 (2022), DOI: 10.1038/s41467-022-28133-x
- 72 Y. Hu, W. Gong, S. Wei, S. Khuje, Y. Huang, Z. Li, Y. C. Li, F. Yao, **Q. Yan**, S. Ren, "Lithiating Magneto-Ionics in a Rechargeable Battery" *PNAS* 119, e2122866119 (2022), DOI: 10.1073/pnas.2122866119
- 71 A. Sheng, S. Khuje, J. Yu, T. Parker, J.-Y. Tsai, L. An, Y. Huang, Z. Li, C.-G. Zhuang, L. Kester, **Q. Yan**, S. Ren, "Copper Nanoplates for Printing Flexible High-Temperature Conductors". *ACS Appl. Nano Mater.* 5, 4028 (2022), DOI: 10.1021/acsanm.2c00019

- 70 H. Bai, P. Chu, J. Y. Tsai, N. Wilson, X. Qian, **Q. Yan**, H. Ling, "Graph Neural Network for Hamiltonian-Based Material Property Prediction" *Neural Comput. Appl.* 34, 4625 (2022), DOI: 10.1007/s00521-021-06616-0
- 69 M. Richter, E. Peterson, L. Zhou, A. Shinde, P. Newhouse, **Q. Yan**, S. Fackler, J. Yano, J. Cooper, K. Persson, J. Neaton, J. Gregoire, "Band Edge Energy Tuning Through Electronic Character Hybridization in Ternary Metal Vanadates" *Chem. Mater.* 33, 7242 (2021), DOI: 10.1021/acs.chemmater.1c01415
- 68 H. R. Banjade, S. Hauri, S. Zhang, F. Ricci, W. Gong, G. Hautier, S. Vucetic, **Q. Yan**, "Structure motif centric learning framework for inorganic crystalline systems", *Sci. Adv.* 7, eabf1754 (2021), DOI: 10.1126/sciadv.abf1754
- 67 W. Gong, **Q. Yan**, "Graph-based deep learning frameworks for molecules and solid-state materials", *Comput. Mater. Sci.* 195, 110332 (2021), DOI: 10.1016/j.commatsci.2021.110332
- 66 Y. F. Zhang, J. Pan, H. Banjade, J. Yu, H. Lin, A. Bansil, S. Du, **Q. Yan**, "Two-dimensional MX Dirac materials and quantum spin Hall insulators with tunable electronic and topological properties", *Nano Res.* 14, 584 (2021), DOI: 10.1007/s12274-020-3022-3
Selected as cover article
- 65 H. R. Banjade, J. Pan, **Q. Yan**, "Monolayer 2D semiconducting tellurides for high-mobility electronics" *Phys. Rev. Mater.* 5, 014005 (2021), DOI: 10.1103/physrevmaterials.5.014005
- 64 N. H. Attanayake, H. R. Banjade, A. C. Thenuwara, B. Anasori, **Q. Yan**, D. R. Strongin, "Electrocatalytic CO₂ reduction on Earth Abundant 2D Mo₂C and Ti₃C₂ MXenes", *Chem. Commun.* 57, 1675 (2021), DOI: 10.1039/d0cc05822j
- 63 F. Hu, S. C. Abeyweera, J. Yu, D. Zhang, Y. Wang, **Q. Yan**, Y. Sun, "Quantifying Electrocatalytic Reduction of CO₂ on Twin Boundaries" *Chem* 6, 3007 (2020), DOI: 10.1016/j.chempr.2020.07.026
- 62 L. Zhang, J. Pan, **Q. Yan**, Z. Hu, "Computational Study of a Novel 2D Ferromagnetic Metal: the Ce₂C Monolayer", *Phys. Status Solidi RRL* 14, 2000324 (2020), DOI: 10.1002/pssr.202000324
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